Minor Project: <u>Exploratory Data Analysis (EDA) on Fashion Retail Sales</u>

❖ **Project Title:**
Fashion Retail Sales Analysis using Exploratory Data Analysis (EDA)

❖ **Objectives:**
- To explore and analyze fashion retail sales data using EDA techniques.
- To identify key patterns in product performance and sales trends.
- To understand how customer ratings and payment methods impact sales.

❖ **Tools and Libraries Used:**

Tools:

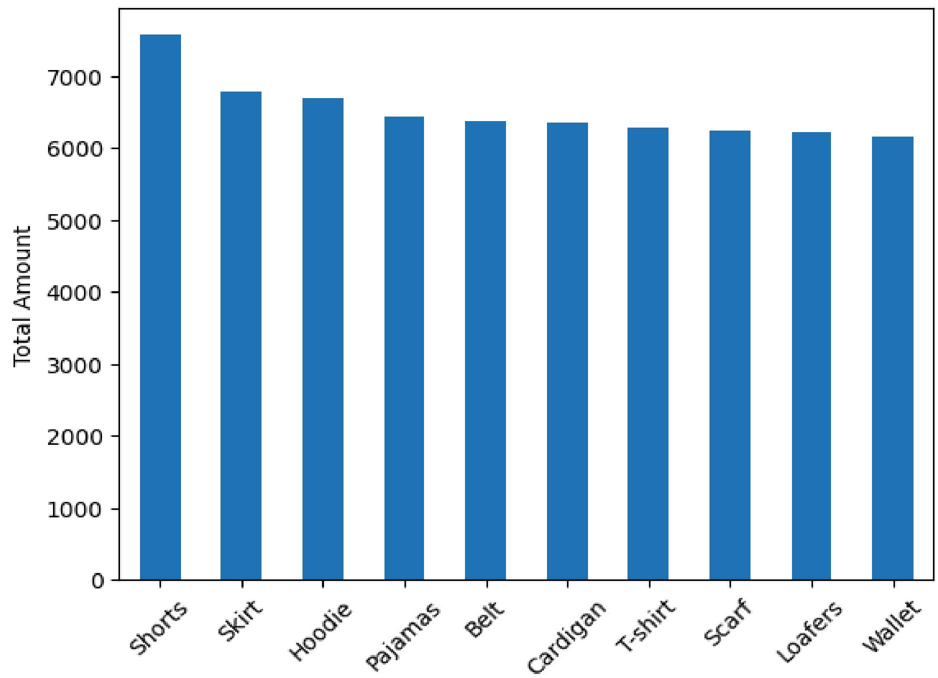
- Google Colab
- Python

Libraries:

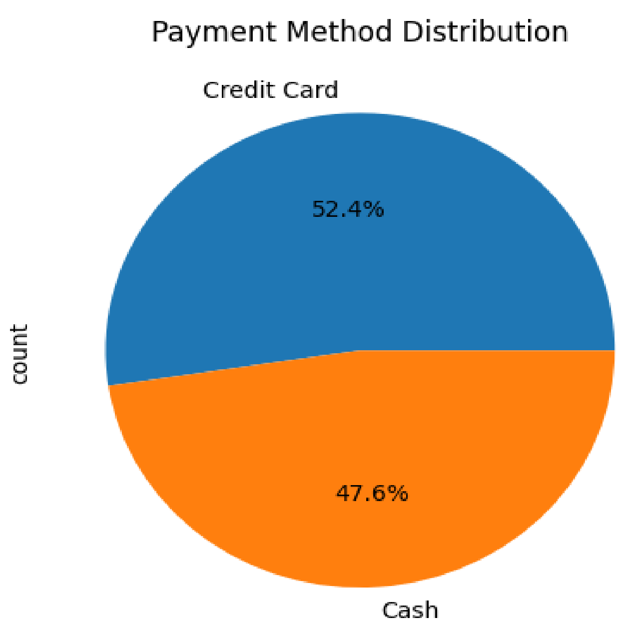
- Pandas
- NumPy
- Matplotlib
- Seaborn

❖ Screenshots of Key Visualizations:

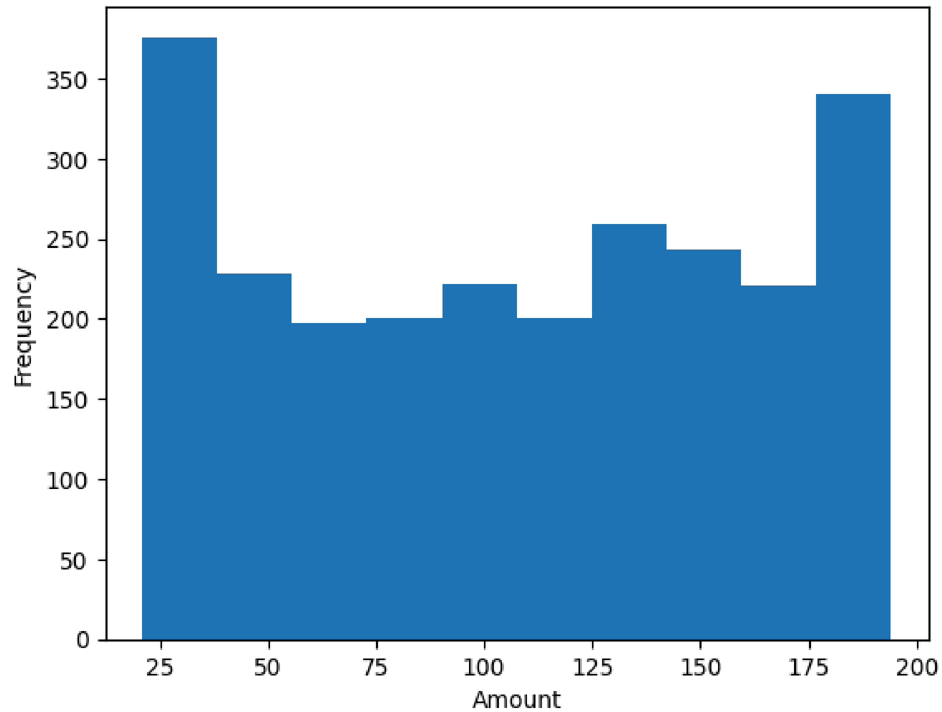
1. Top 15 Items Purchased by Sales



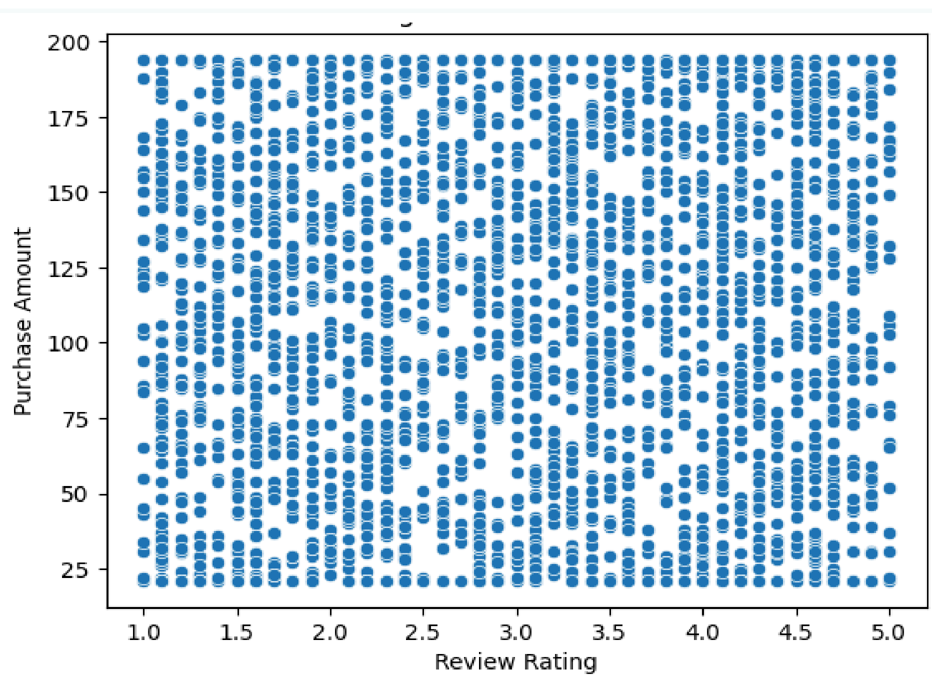
2. Payment Method Distribution



3. Purchase Amount Distribution



4. Rating vs Purchase Amount



❖ Summary of Insights:

- The analysis shows that certain items like shorts, skirts, and hoodies generate the highest sales, indicating strong customer preference for these products.
- The purchase amount distribution is fairly spread out, but most transactions fall within a moderate price range, suggesting consistent spending behavior.
- From the payment method analysis, credit cards are slightly more preferred than cash, showing a growing trend towards digital payments.
- The relationship between review rating and purchase amount does not show a strong direct correlation, indicating that higher spending does not always result in higher ratings.

❖ Challenges Faced & Solutions:

- **Large dataset made analysis complex**
  Solution: Used top values (Top 10/15) to simplify visualizations.
- **Outliers in purchase amoun**t
  Solution: Identified using box plot and handled using capping method.
- **Limited dataset features (no demographic data)**
  Solution: Focused analysis on available columns like items, ratings, and payment methods.
- **Overcrowded visualizations**
  Solution: Selected only important graphs for better clarity.